

Book III - The Spiral Steward
Part 3 - How a Forest Becomes a Forest

Chapter 12 - What Bioinformatics Taught Me About Freedom

BRIDGE 1 — What Bioinformatics Taught Me About Freedom

Type: Personal testimony + systems science **Core Claim:** The discovery that no gene controls all other genes is the same discovery that no person should govern all other people. The science leads directly to the philosophy.

The Personal Hook

This is Sage's origin story for the channel — the moment the biology and the philosophy fused.

Most people arrive at libertarianism or voluntaryism through political philosophy — Locke, Bastiat, Rothbard, Larken Rose.

Sage arrived through a living system.

When you spend enough time studying how genomes actually work — how thousands of genes regulate each other through distributed feedback loops with no master controller — you stop being able to take centralized authority seriously. Not because of a political argument. Because of what you've *seen* in the data.

The biology is the argument.

The Scientific Foundation

Gene Regulatory Networks: A gene regulatory network (GRN) is the system by which genes turn each other on and off. Activators. Repressors. Feedback loops. Feed-forward loops. Thousands of genes, each responding to signals from its neighbors.

No master gene. No gene that sits at the top of the hierarchy and commands all others. The most studied regulatory genes (transcription factors, master regulators) are themselves regulated. Every "controller" is also controlled.

The intelligence of the genome is distributed across the entire network. When researchers try to understand development by studying one gene in isolation — they consistently fail. The behavior is in the network, not in any node.

Single-Cell Analysis: Before single-cell sequencing, biologists thought of tissues as uniform. A liver was a liver. A brain was a brain.

Single-cell RNA sequencing revealed: within what looks like a uniform tissue, there are dozens of distinct cell types — each running the same genome but expressing it differently based on local signals and history.

No two cells are identical. Every cell is specialized. No cell is above another in the hierarchy of value. Each performs a function the whole cannot do without.

This is not a metaphor. This is the literal architecture of a working body.

The Pivot

"When I first saw a single-cell UMAP — thousands of cells, each a dot, clustering by type, no two exactly the same, all running the same code — I thought: this is what the True Republic looks like."

The Living Age Application

The centralized state assumes that the intelligence needed to run society can be gathered into one place — a legislature, a central bank, a planning committee.

This is the biological equivalent of assuming one gene can regulate all others. Every attempt to build such a gene has failed. Every attempt to build such a state has failed. For the same reason: the knowledge is distributed. It cannot be centralized without being destroyed.

Hayek called this the knowledge problem. Biology calls it the limits of master regulators. They are the same problem.

The Living Age solution — voluntary, decentralized, emergent — is not a political preference. It is what happens when you design society the way the genome designs a body: give every node its own agency, let it respond to local signals, protect feedback loops, and let the order emerge.

The LivingWorks Vision

"We need to put together: architects, civil engineering, economics, bioinformatics, computer science, linguistics, AI, philosophy, political science, spontaneous order, Daoism, the Force and hobbits, media. All working together to create a decentralized manufacturing system based on life not death."

Bioinformatics gives us the microscope for this philosophy. Not only seeing life's code, but seeing life's *order* — and learning to design civilization the way life designs itself.

Not a blueprint handed down from above. A set of conditions from which the right form can emerge.

Cold Open:

"Every gene in your body is regulated by other genes. And those genes are regulated by other genes. And it goes all the way down. There is no master gene. There is no top of the hierarchy. And yet — your body works perfectly. So I started asking: what if civilization worked the same way?"

The Science: Gene regulatory networks. Single-cell analysis. No master controller. The intelligence is in the network.

The Pivot: How studying living systems made centralized authority intellectually impossible to take seriously.

The Application: The knowledge problem. The True Republic as a biological model. Personalized medicine → personalized governance.

The Personal: From bioinformatics student to Sagent. The science wasn't separate from the philosophy — it was the philosophy.

The Vision: LivingWorks. The design platform for the Living Age. The next civilization built the way a genome builds a body.

Source Quotes

"People are like cells in a body — I will use that concept and apply the principles of bioinformatics to help guide people toward their individualized choices. Similar to single cell analysis."

"Bioinformatics is the study of how life organizes itself — and that makes it, at its best, a form of stewardship."

"When you do computational biology, you are forced into humility. You learn that you cannot design an entire living world top-down."

"life is not a block you carve into shape; it is a spiral you cultivate."

"I am studying bioinformatics because things are either true or not true."

The Living Age Trilogy